

# **ADDAX AD13A**

## **THREE PHASE METERS**



### **TECHNICAL DESCRIPTION**

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## Revision History

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## Terms and Abbreviations

| Abbreviation | Description                                |
|--------------|--------------------------------------------|
| <b>AMI</b>   | Advanced Metering Infrastructure           |
| <b>CIU</b>   | Customer Interface Unit                    |
| <b>DC</b>    | Direct Connected (meter)                   |
| <b>HHU</b>   | Hand Held Unit                             |
| <b>LCD</b>   | Liquid-Crystal Display                     |
| <b>LED</b>   | Light Emitting Diode                       |
| <b>LV</b>    | Low Voltage                                |
| <b>OFDM</b>  | Orthogonal Frequency Division Multiplexing |
| <b>PLC</b>   | Power Line Communication                   |
| <b>PLM</b>   | Power Line Modem                           |
| <b>RMS</b>   | Root Mean Square                           |
| <b>TC</b>    | Transformer Connected (meter)              |
| <b>THD</b>   | Total Harmonic Distortion                  |
| <b>TOU</b>   | Time of Use                                |

# 1. GENERAL

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Given Technical Description is intended for studying technical characteristics of three phase electricity ADDAX meters based on ADDAX technology.

The Meter can operate independently or as a part of the ADDAX IMS system.

Here, the most complete set of the meter features is described. In specific cases, limited functionality can be implemented in different models/variants.

## 1.1. Purpose

Three-phase meters (hereinafter Meters) are solid state smart devices for active and reactive energy metering. The meters are designed to measure instantaneous power, consumed and generated active and reactive energy in three phase alternating current circuits.

The meters are generally used on the residential market. The focus of ADDAX IMS is AMR/AMI basic features for deregulated electricity market: TOU tariffs, load profiles, energy quality control etc.

The meter optionally supports the following communication interfaces for various purposes, such as:

1. Power Line Communication (PLC) based on OFDM communication solution.
2. RS-485 interfaces (according to EIA/TIA-485A standard) – as an auxiliary interface for connection of extension modules.
3. M-Bus wireless interfaces for HAN communication.

## 1.2. Key Features

|                                                                                                                     |                                                                                                                                                                          |
|---------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Interoperability                                                                                                    | Provides functional and informational meter compatibility with standard solutions of other metering equipment and software manufactures                                  |
| Multi-functional                                                                                                    | Provides functionality of smart meters as a part of AMI compliant solution                                                                                               |
| Multi-tariff registration                                                                                           | Allows flexible TOU tariff structure to decrease peak loads and ensure the total consumption balancing based on consumption interval data                                |
| 2-way communication                                                                                                 | Allows the Meter integration to advanced metering system                                                                                                                 |
| Supercapacitor/Battery                                                                                              | Backup power source supports clock/meter operation when the power is off                                                                                                 |
| Meter cover and terminal block cover opening sensors, magnetic field sensor                                         | Controls theft or fraud attempts                                                                                                                                         |
| Built-in basic relays in phase wire (and optionally in neutral wire, depending on the meter model) and extra relays | Provide effective energy consumption control as well as demand side management features                                                                                  |
| Additional communication interfaces (RS-485, M-Bus wireless) (optionally, depending on meter model)                 | Provide channels for connection of various communication modules for WAN communication (e.g. 3GPP), for connection with other devices and channels for HAN communication |

### 1.3. Standards

The Meter meets the requirements of the following international standards:

|                     |                                                                                                                                                            |
|---------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------|
| IEC 61010-1:2001-02 | Safety requirements for electrical equipment for measurement, control and laboratory use. Part 1. General requirements                                     |
| IEC 62052-11:2003   | Electricity metering equipment (AC) - General requirements, tests and test conditions - Part 11: Metering equipment<br>Maintenance Result Date: 2012-02-01 |
| IEC 62052-21        | Electricity metering equipment (a.c.) – General requirements, tests and test conditions – Part 21: Tariff and load control equipment                       |
| IEC 62053-21:2005   | Electricity metering equipment (a.c.) – Particular requirements - Part 21: Static meters for active energy (classes 1 and 2)                               |
| EN 62053-23         | Electricity measuring equipment (a. c), Particular requirements Part 23: Static meters for reactive energy (class indexes 2 and 3)                         |
| IEC 62053-22        | Electricity metering equipment (a.c.) – Particular requirements - Part 22: Static meters for active energy (classes 0,2 S and 0,5 S)                       |
| EN 50470-1:2005     | Electricity metering equipment (a.c.) Part 1: General requirements, test and test conditions-Metering equipment (class indexes A, B and C)                 |
| EN 50470-3:2005     | Electricity metering equipment (a.c.) Part 3: Particular requirements – Static meters for active energy (class indexes A, B and C)                         |
| IEC 62056-61:2002   | Electricity metering – Data exchange for meter reading, tariff and load control – Part 61: Object Identification System OBIS                               |
| IEC 62054-21        | Electricity metering (a.c.) Tariff and load control Part 21: Particular requirements for time switches                                                     |
| G3 specifications   | ITU-T_G.9903-201210. ITU-T Transmission Systems And Media, Digital Systems And Networks Access Networks – In Premises Networks                             |
| IEC 62056-21:2002   | Electricity metering – Data exchange for meter reading, tariff and load control – Part 21: Direct local data exchange                                      |
| EN 62053-52         | Electricity metering equipment (a.c.) – Particular requirements – Part 52: Symbols (IEC 62053-52)                                                          |
| IEC 62053-31        | Electricity Metering Equipment (a.c.) - Particular Requirements – Pulse output devices for electromechanical and electronic meters (two wires only)        |
| IEC 61038           | Time switches for tariff and load control                                                                                                                  |
| IEC 62053-61        | Electricity Metering Equipment (a.c.) - Particular Requirements – Power consumption and voltage requirements                                               |
| IEC 61107           | Data Exchange for Meter Reading, Tariff and Load Control                                                                                                   |
| IEC 60529           | Degrees of protection provided by enclosures. IP code                                                                                                      |
| EN 13757-2          | Communication systems for and remote reading of meters” – Part 2: Physical and link layer                                                                  |
| EN 13757-3          | Communication systems for and remote reading of meters “ – Part 3: Dedicated application layer                                                             |

|                                      |                                                                                                                                                                                                                                                                                                                           |
|--------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| EN 13757-4                           | Communication systems for meters and remote reading of meters.<br>Wireless meter readout (radio meter reading for operation in the 868-870 MHz SRD band)                                                                                                                                                                  |
| CENELEC<br>EN 50065-1:2011           | Signalling on low-voltage electrical installations in the frequency range 3 kHz to 148,5 kHz - Part 1: General requirements, frequency bands and electromagnetic disturbances                                                                                                                                             |
| CENELEC<br>EN 50065-2-1:2003/A1:2005 | Signalling on low-voltage electrical installations in the frequency range 3 kHz to 148,5 kHz - Part 2-1: Immunity requirements for mains communications equipment and systems operating in the range of frequencies 95 kHz to 148,5 kHz and intended for use in residential, commercial and light industrial environments |
| CENELEC<br>EN 50065-7:2001           | Signalling on low-voltage electrical installations in the frequency range 3 kHz to 148,5 kHz — Part 7: Equipment impedance                                                                                                                                                                                                |
| IEC 62056-7-5<br>(coming soon)       | Electricity Metering Data Exchange – The DLMS/COSEM Suite – Part 7-5: Local data transmission profiles for Local Networks (LN).                                                                                                                                                                                           |

## 2. SPECIFICATIONS

### ○ Meter Designation and Modifications

The following system is used to indicate the meter models within the family:

| ADDAX                                                                                                                          | AD1X | Y. | Z- | V-W |
|--------------------------------------------------------------------------------------------------------------------------------|------|----|----|-----|
| ADD GRUP                                                                                                                       |      |    |    |     |
| <b>mseries:</b><br>X = 1: single-phase meter<br>X = 2: three-phase meter                                                       |      |    |    |     |
| <b>Case Type:</b><br>Y = A, B or C: classic meters<br>Y = S: split meters                                                      |      |    |    |     |
| <b>Production series indicator:</b> (max. current, max. voltage and type of connection)<br>Z <sub>max</sub> Z <sub>max</sub> Z |      |    |    |     |
| <b>Ordering information:</b><br>V: Assembly index<br>W: Version Index                                                          |      |    |    |     |

The meter design features are presented below:

1. Relays in phase wire (DC meters only);
2. Extra relays;
3. Meter cover opening sensor;
4. Terminal block cover opening sensor;
5. Magnetic field sensor;
6. Backlit display;
7. Supercapacitor and/or battery.

Available communications:

8. PRIME (optionally);
9. G3-PLC (optionally);
10. RS-485 (optionally);
11. Wireless M-Bus (optionally).

## ○ Technical Specifications

The meters base technical characteristics are presented in the table below.

**Table 2.2.1.** Technical specifications

| Parameter                                                 | Value                                                                                         |
|-----------------------------------------------------------|-----------------------------------------------------------------------------------------------|
| Reference voltage, $U_n$<br>(depending on meter model)    | 3×57.7/100 V (TC meters only); 3×120/208 V;<br>3×230/400 V;<br>3×240/415 V (ANSI meters only) |
| Voltage range                                             | 0.8 $U_n$ ... 1.2 $U_n$                                                                       |
| Reference frequency<br>(depending on meter model)         | 50 Hz ( $\pm 2\%$ )<br>60 Hz ( $\pm 2\%$ )                                                    |
| Reference current<br>(depending on meter model) $I_{ref}$ | 5/10 A – DC meters<br>2.5/5 A – TC meters                                                     |
| Maximum current<br>(depending on meter model)             | 80/100 A – DC meters<br>10 A – TC meters                                                      |
| Minimum current<br>(depending on meter model)             |                                                                                               |
| Active energy                                             | 0.05 $I_{ref}$ – DC meters<br>0.01 $I_{ref}$ – TC meters                                      |
| Reactive energy                                           | 0.05 $I_{ref}$ – DC meters<br>0.02 $I_{ref}$ – TC meters                                      |
| Starting current<br>(depending on meter model)            |                                                                                               |
| Active energy measurement                                 | 0.004 $I_{ref}$ – DC meters<br>0.001 $I_{ref}$ – TC meters                                    |
| Reactive energy measurement                               | 0.005 $I_{ref}$ – DC meters<br>0.003 $I_{ref}$ – TC meters                                    |
| Accuracy class<br>(depending on meter model)              |                                                                                               |
| Active energy<br>(EN 50470-3, IEC 62053-21)               | B (1) – DC meters<br>C (0.5) – TC meters                                                      |
| Reactive energy<br>(IEC 62053-23)                         | 2 – DC/TC meters                                                                              |
| Meter constant<br>(depending on meter model)              |                                                                                               |
| Active energy                                             | 1000/2000 imp/kWh – DC meters<br>10 000/20 000/50 000* imp/kWh – TC meters                    |
| Reactive energy                                           | 1000/2000 imp/kArh – DC meters                                                                |

\* For the transformer connected meters with nominal voltage of 57.7 V.

\*\* Meters with RS-485-2 interface have IP 51.

| Parameter                                                             | Value                                                                                                   |
|-----------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                                                                       | 10 000/20 000/50 000* imp/kvArh - TC meters                                                             |
| PL communication interface (optional)                                 | PL (LV 0,4 kV)<br>PRIME/G3-PLC modulation                                                               |
|                                                                       | Data transmission rates up to 21-64 kbps (PRIME) / 20-46 kbps (G3-PLC) with coding (on physical layer)  |
| Local communication interface                                         | Optical port                                                                                            |
| Internal clock                                                        | quartz crystal 32 kHz                                                                                   |
| Clock accuracy (at 23°C)                                              | ≤0.5 s /24 h                                                                                            |
| Operation temperature range (IEC 60721-3-3)                           | -40°C ... +70°C                                                                                         |
| Storage temperature (IEC 60721-3-1)                                   | -40°C ... +70°C                                                                                         |
| Relative humidity (at 23°C, without moisture condensation)            | 95 %                                                                                                    |
| Inherent consumption (IEC 62052-11, IEC 62053/21-22-23, IEC 62053-61) |                                                                                                         |
| of current circuit, not more than                                     | 2.5 VA                                                                                                  |
| of voltage circuit, not more than                                     | 2 W/10 VA per phase                                                                                     |
| with connection to communication interface, not more than             | 3 W/10 VA per phase                                                                                     |
| Insulation strength (IEC 61010-1-90)                                  | 4 kV, 50 Hz, 1 min                                                                                      |
| Shock voltage (IEC 62052-11)                                          | 8 kV<br>pulse waveform 1.2/50 µs according to (IEC 60060-1);<br>the source impedance: 500 ohms ±50 ohms |
| Impulse voltage withstand (SP-method 1618; EN 50470-1)                | 12 kV,<br>pulse waveform 1.2/50 µs according to (IEC 60060-1)                                           |
| Electrostatic discharge (EN 61000-4-2)                                | 15 kV                                                                                                   |
| High frequency radiant field (IEC 61000-4-3)                          | 10 V/m                                                                                                  |
| Immunity against HF electric fields 80 - 2000 MHz (IEC 61000-4-3)     | ≥ 30 V/m                                                                                                |
| High frequency interferences (IEC 61000-4-4)                          | 4 kV                                                                                                    |
| MTBF, less than, per year                                             | 0.5 %                                                                                                   |

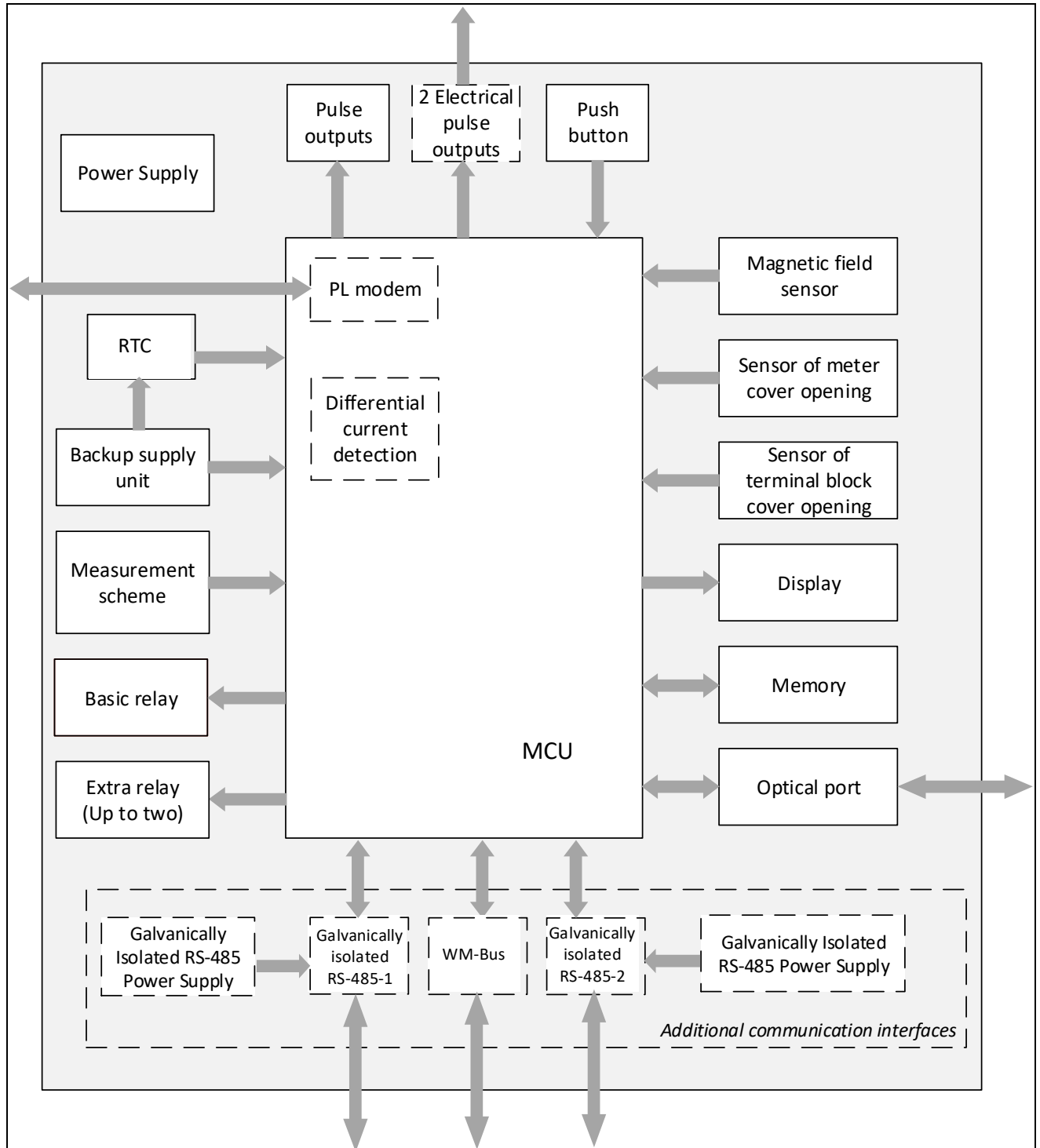
| Parameter                    | Value             |
|------------------------------|-------------------|
| Mean lifetime, not less than | 20 years          |
| IP rating (EN 60529)         | IP54**            |
| Dimensions                   | 280 x 180 x 68 mm |

### 3. METER DESIGN

#### ○ Meter Block-Diagram

The meter block diagram is presented in the [Fig 3.1.1.](#)

**Note:** Units, marked with dotted lines, are optional and are available according to the meter type.



**Fig. 3.1.1.** Meter block diagram

### 3.1. Meter Basic Units

|                                                          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|----------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Metering Unit</i></b>                              | <ul style="list-style-type: none"> <li>✓ Main components - voltage and current sensors.</li> <li>✓ Precision dividers to measure voltage are used. Dividers diminish input voltage up to convenient for metering scheme value. Voltage division is performed with optimal linearity at minimal phase shift.</li> <li>✓ Transformers for measuring phase and neutral (DC meters only) currents are used.</li> <li>✓ Measuring of RMS voltage, RMS current, neutral current (DC meters only), mains period, active energy in both directions, 4 quadrant reactive energy, apparent energy, active power, reactive power, apparent power, angle between current and voltage per phase, angles between voltages, and temperature and converting obtained results to the digital code.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                               |
| <b><i>MCU (Meter Controller Unit)</i></b>                | <ul style="list-style-type: none"> <li>✓ Meter calculates such parameters, as frequency, voltage and current harmonics (up to 13th), THD for voltage and current, voltage imbalance, total active energy, total active power and total reactive power.</li> <li>✓ Storage in the non-volatile memory of measuring data, calibration factors and configuration as well as working firmware.</li> <li>✓ Real time clock support.</li> <li>✓ Support of communication via the local optical port.</li> <li>✓ Data exchange with PLC modem and extension communication module (if any).</li> <li>✓ Information displaying.</li> <li>✓ Generation of signals to the test pulse outputs (active and reactive energy) and electrical pulse outputs (active and reactive energy and time synchronization) (if any).</li> <li>✓ Switching-off relay control (via the basic relay, if any). The meter continuously monitors the physical state of relay and registers relevant events in special logs.</li> <li>✓ Secondary load control (via the extra relays).</li> <li>✓ Terminal block and meter case opening registration.</li> <li>✓ Magnetic field sensor control.</li> <li>✓ Push button control.</li> </ul> |
| <b><i>Power Supply, including Backup Supply Unit</i></b> | <ul style="list-style-type: none"> <li>- Forms voltages necessary to supply meter functional units: provides the normal operation mode in the range of supply-line voltages of 85 – 275 V.</li> <li>- Supports supercapacitor to sustain RTC for not less than 120 h, when the power is off (optionally)</li> <li>- Supports a battery as backup supply unit: 20 years lifetime and not less than 3 years of continuous operation.</li> <li>- In energy saving mode the battery provides: <ul style="list-style-type: none"> <li>• operating of meter case and terminal block cover opening sensors;</li> <li>• possibility to fix time of opening sensors actuation;</li> <li>• data displaying;</li> <li>• data exchange via optical port.</li> </ul> </li> <li>- After power restoration the meter automatically goes to normal operation mode.</li> </ul>                                                                                                                                                                                                                                                                                                                                              |

**Communication Unit**

**PLC module (optional)**

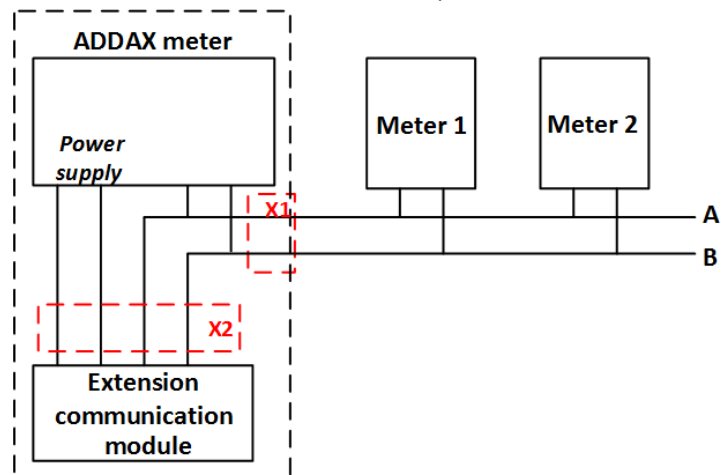
- PLC Modem ensures data exchange via low voltage AC power line.
- The PLM is built-in and comprises:
  - Coupler – a unit to couple with 0.4 kV Power Line
  - PLC modem unit.
  - Zero crossing detection module.

**Additional communication interfaces (optional)**

**RS-485-1**

RS-485 interface supports the following features:

- Meets the requirements of the EIA/TIA-485A standard;
- Two outputs are located within the meter terminal block;
- Includes 2 paralleled outputs:
  1. **X1** – allows the connection of multiple meters to a single RS-485 bus (e.g.: to ensure the connection between the meter with a 3GPP extension communication module and some other meters (without 3GPP communication) to provide data transmission to HES via the same communication channel);
  2. **X2** – provides a link between the meter and an extension communication module and ensures the power supply of the extension communication module;



- Protection against short current and overvoltage is ensured;
- Baud rate - up to 256 kbps;
- Several slave meters can be connected simultaneously. Each meter on the bus must have a unique physical address.
- In energy saving mode the RS-485 channel is not active.

**Wireless M-Bus (WM-Bus)**

To provide data exchange with other intellectual devices on consumer's side (e.g. Customer Interface Unit (CIU)) and ensure data collection and management for HAN, the wireless M-Bus communication interface is used (optional, depending on the meter model).

WM-Bus interface supports the following features:

- Meets the requirements of the EN 13757-4 standard;
- Baud rate - 32 kbps;
- Data transmission distance – up to 150 m;

|                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                           | <ul style="list-style-type: none"> <li>- Frequency – 868 Mhz;</li> <li>- Placed on-board and features an internal antenna;</li> </ul> <p>In energy saving mode the WM-Bus channel is not active. <b><u>RS-485-2 Customer interface</u></b></p> <p>To provide data transmission to HAN devices (e.g. CIU) (optional, depending on the meter model).</p> <p>RS-485-2 interface supports the following features:</p> <p>Meets the requirements of the EIA/TIA-485A standard;</p> <p>Baud rate - up to 256 kbps;</p> <p>Placed on-board and respective connector is provided on the meter case cover;</p> <p>In energy saving mode the RS-485-2 channel is not active.</p>                                                                                                                                                                                                                                                                           |
|                                           | <p align="center"><b>Optical port</b></p> <ul style="list-style-type: none"> <li>- Meets the requirements of IEC 62056-21-2002 and provides data transmission rate up to 115200 bps. The optical port may be accessed by any portable and mobile device (handheld device, PC, laptop, etc.), compliant with specified requirements, using an IR optical head.</li> <li>- The optical head for accessing the port, while in operation, is fixed magnetically on meter front panel and does not influence meter normal operation.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                       |
| <b>Sensors</b>                            | <ul style="list-style-type: none"> <li>- Meter cover and terminal block cover opening sensors - intended to register the corresponding fraud attempts.</li> <li>- Magnetic field sensor allows detecting of extraneous magnetic field and registering the relevant alarm in Fraud detection event log, when an extraneous magnetic field of 400 mT and more is detected.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Tamper Detecting Scheme (optional)</b> | <ul style="list-style-type: none"> <li>- Differential current is used as an indicator of tampering attempts in case of exceeding the defined value (for DC meters only).</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| <b>Load Control Unit (optional)</b>       | <ul style="list-style-type: none"> <li>- The main component of this unit is the disconnecting relay (80A or 100A) in the phase wire.</li> <li>- The meter can be additionally equipped with a relay in the neutral wire (optionally, for DC meters only).</li> <li>- Base parameters: <ul style="list-style-type: none"> <li>• Relay type - polarized bi-stable relay;</li> <li>• Switching current -80 A (100 A);</li> <li>• Switching voltage -250 V;</li> <li>• Maximum switching voltage – 440 V.</li> </ul> </li> <li>- The relay is managed by the controller. Consumer connection is implemented manually by push button or automatically by time-out (see details in <a href="#">Reference 1</a>).</li> <li>- If a customer was disconnected (e.g. for debts) the relay can be reconnected by a remote command. In this case, it's necessary to disconnect and reconnect a main breaker on entering the customer's apartment.</li> </ul> |

|                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <p align="center"><b>Secondary<br/>Load Control<br/>Unit<br/>(optional)</b></p> | <ul style="list-style-type: none"> <li>- The main components of this unit are 2 low power switching off extra relays (each of 2.5 A).</li> <li>- Relays control external contactors of consumer's appliances to provide Demand Side Management (DSM) functionality.</li> <li>- Base parameters:<br/>Relay type – non-polarized relay;<br/>Switching current - 2.5 A (5 A);<br/>Switching voltage - 250 V;<br/>Maximum switching voltage - 250 V.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| <p align="center"><b>Display</b></p>                                            | <ul style="list-style-type: none"> <li>- Liquid crystal display with backlight is supported.</li> <li>- Displays data in automatic and manual mode.</li> <li>- 8 digits, h =9.2 mm to display metering data.</li> <li>- Special symbols, data identification according to IEC 62056-61 (OBIS codes).</li> </ul> <p>For more details see <a href="#">section 5</a>.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |
| <p align="center"><b>Test Pulse<br/>Output</b></p>                              | <ul style="list-style-type: none"> <li>- The meter is equipped with signal LEDs (for active and reactive energy) on the front panel.</li> <li>- Signal wavelength is 650 nm.</li> <li>- LEDs are blinking in proportion to power consumption as indicated in the <a href="#">table 2.2.1</a>.</li> <li>- Pulse output allows meter metrological parameters control during calibration.</li> <li>- When there is no load the LEDs continuously glow (optionally, depending on the meter firmware).</li> <li>- Depending on meter model, it can support two electrical pulse outputs, placed within the meter terminal block cover (see <a href="#">Fig. 3.5.2</a> and <a href="#">3.5.3</a>). These can be used for transmission of pulses to external pulse receivers. Pulses carry information about the measured parameters. Pulse outputs can be configured for different parameters, such as active energy, reactive energy, and time synchronization.</li> <li>- The main characteristics of electrical pulse outputs are as follows:             <ol style="list-style-type: none"> <li>1. Outputs are electrically isolated by opto-couplers;</li> <li>2. Pulse output complies to Class A IEC 62053-31:                 <ol style="list-style-type: none"> <li>1. Maximum voltage (U<sub>max</sub>) - 27 V DC;</li> <li>2. Maximum current in ON-state - 27 mA;</li> <li>3. Minimum current in ON-state - 10 mA;</li> <li>4. Maximum current in OFF-state -2 mA.</li> </ol> </li> <li>3. Pulse constant is programmed. By default the pulse constant is 500 imp/kWh (imp/kVAh) for direct connected meters, 5 000 imp/kWh (imp/kVAh) for transformer operated meters.</li> </ol> </li> </ul> |

|                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|-------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b><i>Real Time Clock</i></b> | <ul style="list-style-type: none"> <li>- Built-in Real Time Clock (RTC) provides timestamp for metering data and events, support of TOU tariff metering, process control commands according to the set schedule.</li> <li>- Accuracy 0.5 sec per day under standard conditions (<math>t = 25\text{ }^{\circ}\text{C}</math>) according to IEC 62052-21.</li> <li>- During meter operation as a part of the metering system, constant external synchronization of the meter clock with the system clock via data transfer network is ensured.</li> <li>- Local setting and synchronization of the clock is also possible using meter optical port.</li> <li>- Supercapacitor backup supply ensures the clock operation when the power is off.</li> </ul> |
| <b><i>Push Button</i></b>     | <p>The push button is used for different purposes:</p> <ul style="list-style-type: none"> <li>✓ To manually reconnect the customer's load;</li> <li>✓ To scroll the meter screens;</li> <li>✓ To view on data LCD when the power is off.</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

### **3.2. Meter Available Accessories**

The following aids can be used during the meter operation and service:

- ✓ Hand Held Unit (HHU) and IR optical probe can be used for local data exchange;
- ✓ Software for local or remote parameterization using PC.

### **3.3. Meter Design Features**

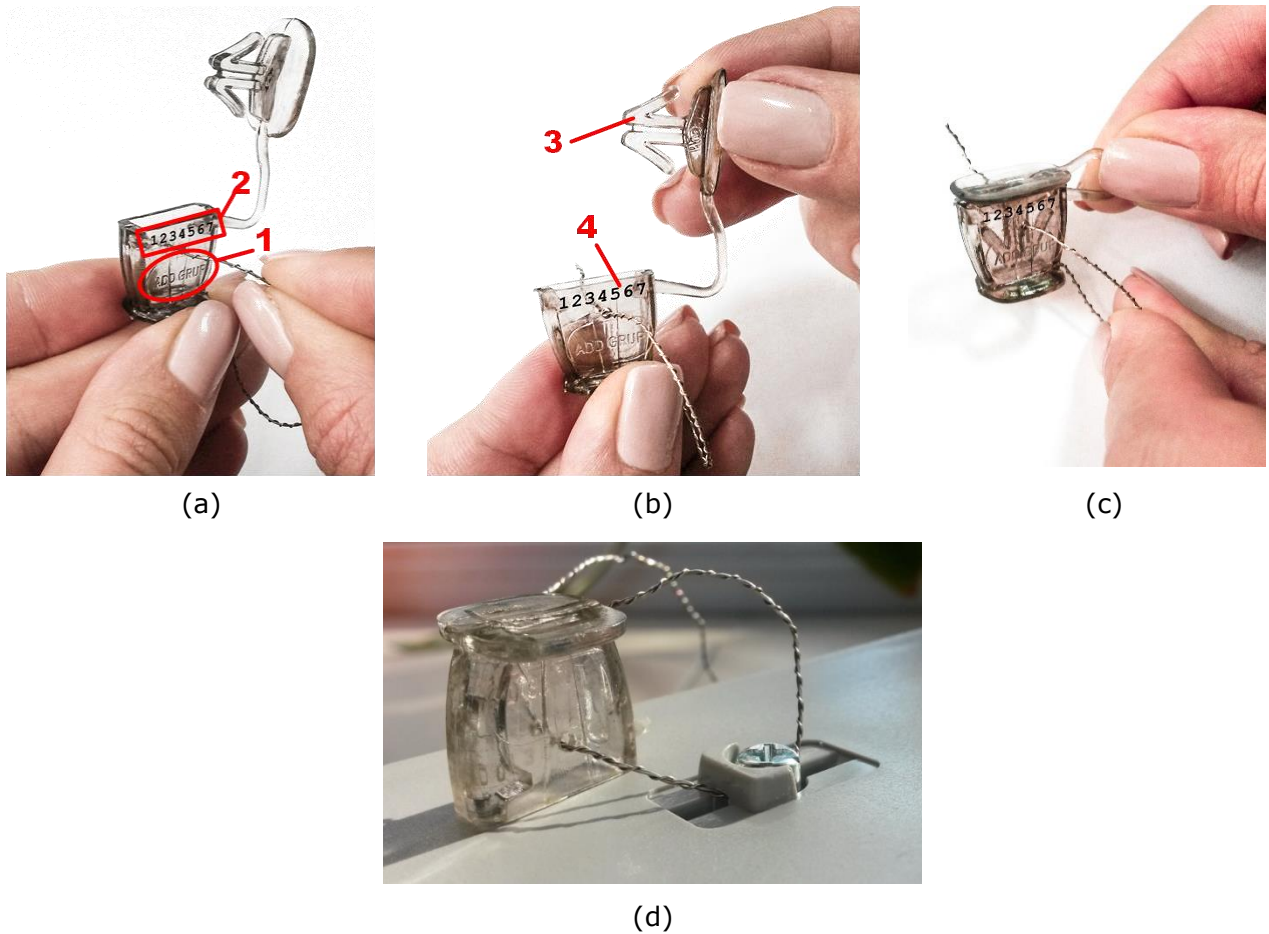
The meter is placed in a case – rectangular plastic box with mounting elements (see [Fig. 3.4.2.](#)). The materials of the meter case and cover meet international standards on mechanical requirements.

Metal parts of the meter are made from stainless metal. In normal operating conditions, all parts of the meter are efficiently protected against corrosion. Electronic components are protected with a coating against water, dust and vermin. The protective layer is strong enough, so that the meter cannot be damaged by weather conditions, under certain operating conditions.

Protection class against water and dust is IP54 – for meters with classic cover, and IP51 – for meters with RS-485-2 interface output located on the meter case cover. The meter case is made from impact-resistant self-extinguishing polycarbonate, meter cover - transparent impact-resistant self-extinguishing polycarbonate. Meter case withstands temperatures of 100°C and more. RS-485-2 interface output is protected with a fire resistant silicone rubber cover (see [Fig. 3.4.2., pos. 6](#)).

The meter cover and terminal block cover are fixed with screws, which can be sealed. The single-use security seal is made of high strength transparent engineering plastic (polycarbonate of various colors and shades) and noncorrosive non-magnetic stainless steel sealing wire. Security seal is immune to any aggressive substances and environment conditions. The meter manufacturer name (ADD GRUP, pos. **1** in the [Fig. 3.4.1.a](#)) and a unique security seal number (consisting of 7 characters of black color, pos. **2** in the [Fig. 3.4.1.a](#)) are written on the security seal and are clearly visible. The security seal itself consists of 2 parts: the top part (anchor, pos. 3 in the [Fig. 3.4.1.b](#)) and the bottom part (casing, pos. 4 in the [Fig 3.4.1.b](#)). Anchor latches and wire are securely fixed into the casing capsule ([see Fig. 3.4.1.c](#)). Any attempt to pull out the wire or to break the security seal in order to open the cover can be easily observed. The terminal cover provides a hole to thread the

security seal wire (see [Fig. 3.4.1.d](#)). The meter design excludes possibility of touching meter terminals with security cord while sealing the cover.



**Fig. 3.4.1. Meter security seal design**

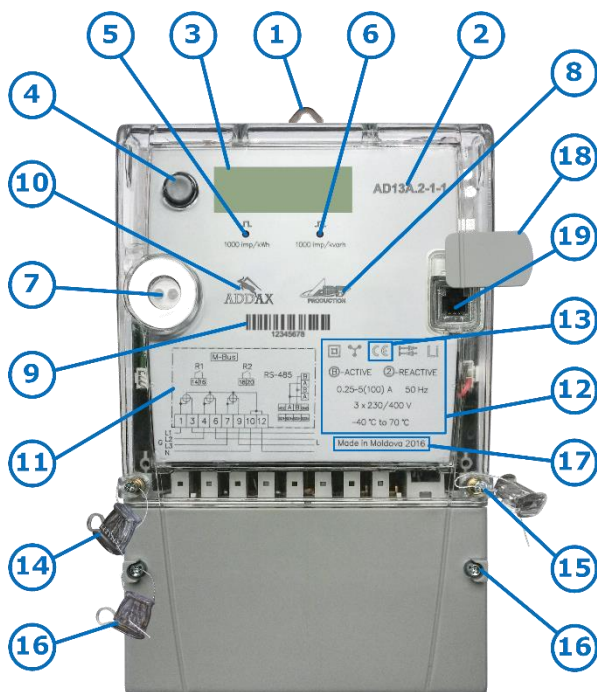
The security seal has the following characteristics:

1. Temperature range: -40 °C to +60 °C;
2. Warranty period: at least 5 years;
3. Service life: at least 16 years.

The meter has facilities for mounting on DIN-rail or in 3 points with/without a fixing bar (see [Fig. 4.2.](#)).

In the [Fig. 3.4.2.](#) the general form of the meter with classical case is showed. In specific models/variants some structural elements may be implemented with restrictions or, on the contrary, extended. When meter features RS-485-2 interface, case cover design differs as proper output is located on the meter cover.

A nameplate with basic meter parameters is placed on the meter case cover. On customer request any additional information, such as description of the OBIS codes, QR code, customer logo and another, can be depicted on the nameplate.

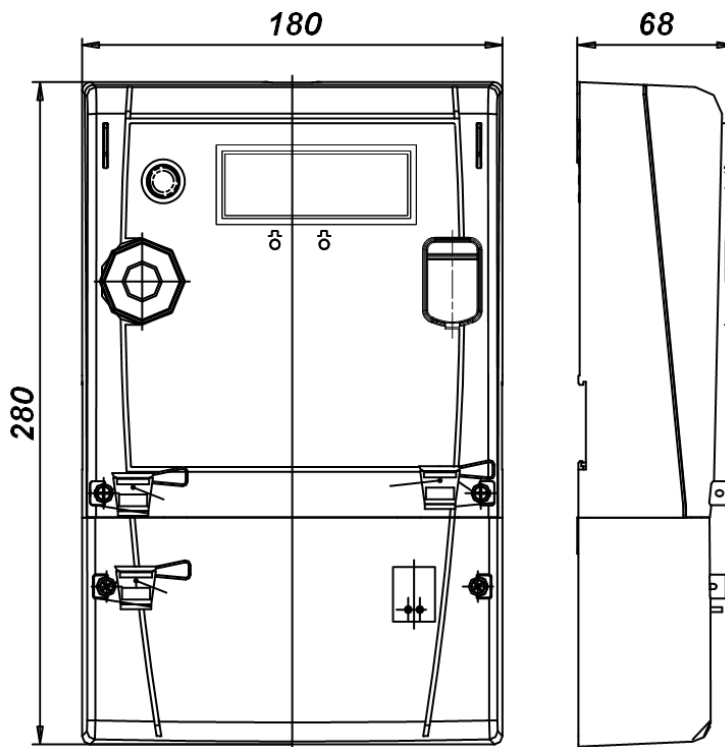


**Fig. 3.4.2.** Meter general form and structural elements

**Table 3.4.1** List of meter structural elements

| Position | Description                                                                                                      |
|----------|------------------------------------------------------------------------------------------------------------------|
| 1        | Button                                                                                                           |
| 2        | Liquid crystal display                                                                                           |
| 3        | Signal LED of active energy                                                                                      |
| 4        | Signal LED of reactive energy                                                                                    |
| 5        | Meter type                                                                                                       |
| 6        | RS-485-2 output cover (optional)                                                                                 |
| 7        | RS-485-2 output (optional)                                                                                       |
| 8        | CE marking, number of EC type examination certificate and supplementary metrology mark "M xx"                    |
| 9        | Seal of metrological service                                                                                     |
| 10       | Seal of power authority                                                                                          |
| 11       | Seal of manufacturer                                                                                             |
| 12       | Connection diagram                                                                                               |
| 13       | Bar-code and meter serial number (the structure and sequence of factory numbers can be agreed with the customer) |
| 14       | Optical port                                                                                                     |

The [Fig.3.4.3.](#) presents overall dimensions of the meter. Meter overall and mounting dimensions comply with the DIN 43857.



**Fig. 3.4.3.** Meter general form and overall dimensions

### ○ Terminal Block

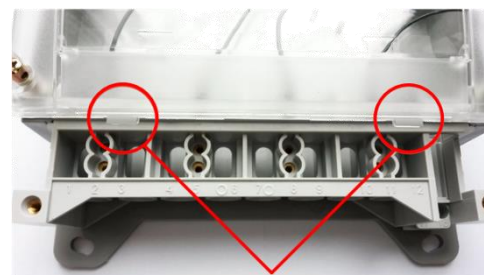
Terminal block is manufactured from impact-resistant, fireproof plastic. Terminal block is closed with opaque plastic cover, screws of which can be sealed. Using terminals with standard parameters allows easy meter installation. Diameter of the terminals for DC meters is 8.5 mm, for TC meters – 5 mm. Screws, fixing the conductors inside the terminals provide reliable and secure connection. Cross-section of the incoming conductors should be selected based on meter maximum current.

The terminal block cover has no cutouts which makes accessible contact parts and wire connection points

The meter terminal block cover is designed so that the meter cover can only be removed after terminal cover has been removed: 2 ribs on the meter case and respectively 2 cuts on the terminal block cover are used (see [Fig. 3.5.1.](#)).

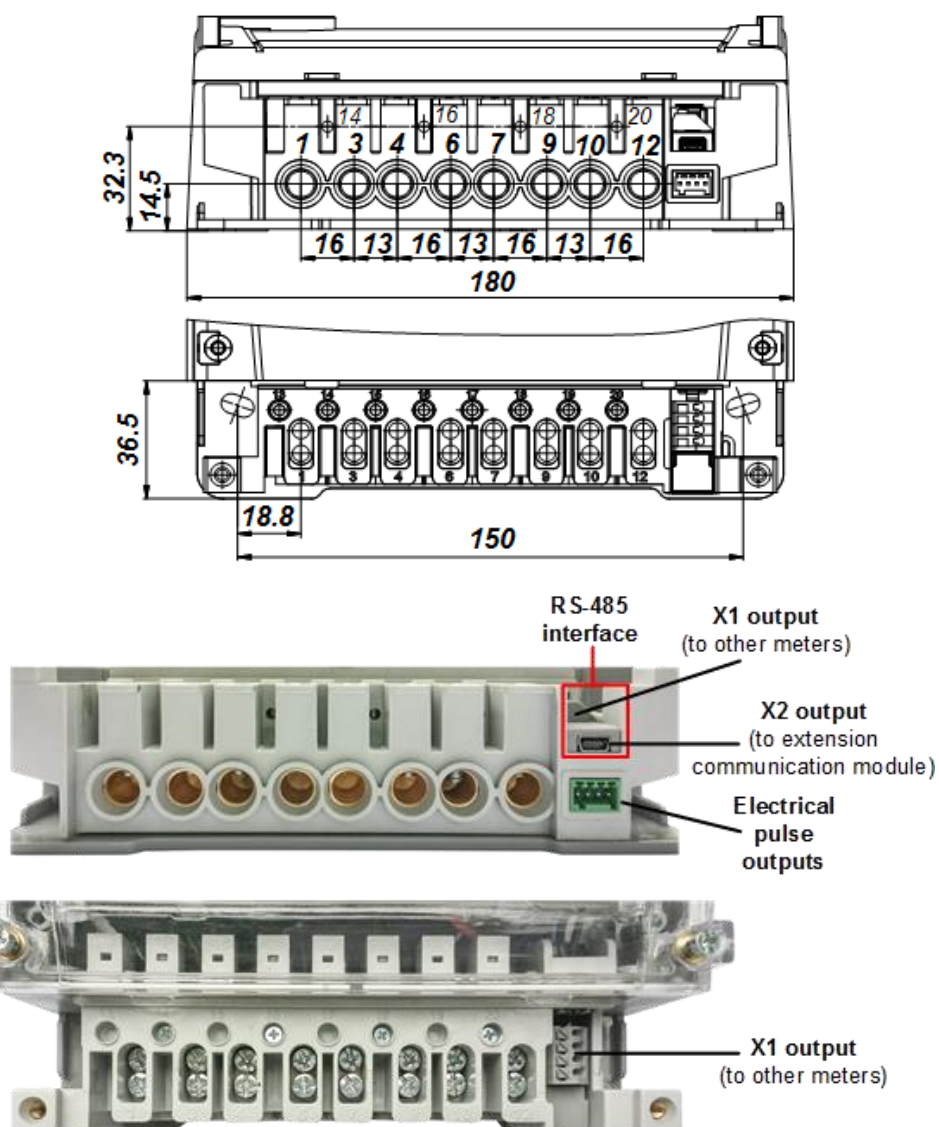


**2 cuts on the terminal  
block cover**

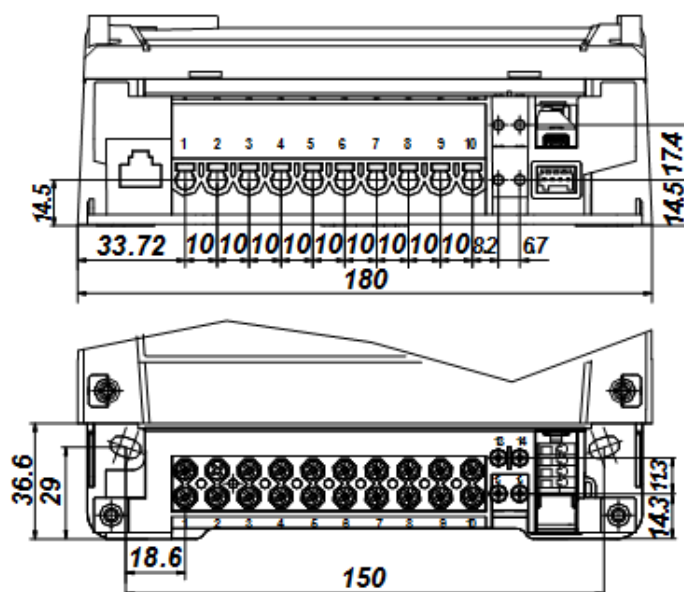


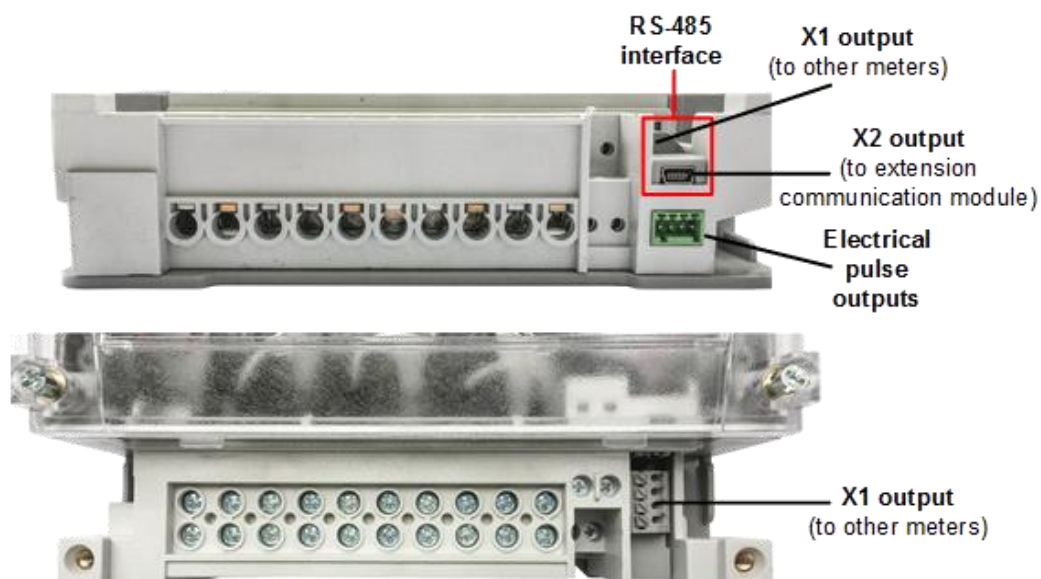
**2 ribs on the meter  
case cover**

**Fig. 3.5.1.** Anti-tampering design features of the meter case and terminal block covers



**Fig. 3.5.2.** Dimensions of terminal block and placement of additional interfaces for DC meter





**Fig. 3.5.3.** Dimensions of terminal block and placement of additional interfaces for TC meter

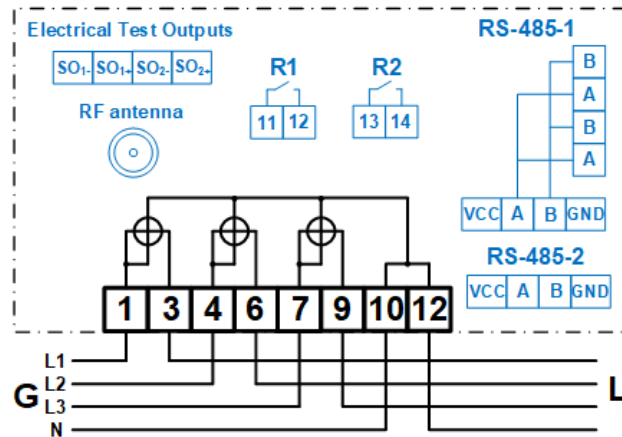
**Table 3.5.1.** Connection of three-phase meter terminals

| Terminals description              | IEC connection | ANSI connection |
|------------------------------------|----------------|-----------------|
| Wires terminals from network side  | 1, 4, 7, 10    | 1, 3, 4, 6      |
| Wires terminals from consumer side | 3, 6, 9, 12    | 7, 9, 10, 12    |
| Neutral wires terminals            | 10, 12         | 6, 7            |

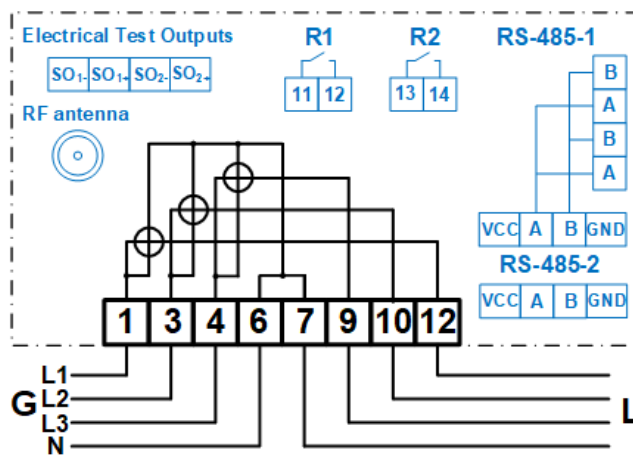
Depending on the meter model extension communication module can be placed under the terminal block cover (e.g. 3GPP). RS-485 interface is used to interconnect the module and meter at this. To indicate 3GPP communication channel status two RGB LEDs can be placed on the meter terminal block cover. Meter design provides easy replacement of the communication module under the terminal block cover.

## 4. METER PUTTING INTO OPERATION

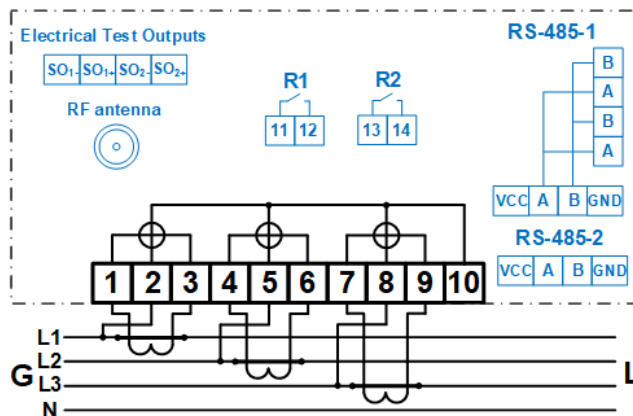
Meter is installed to selected metering point according to connection diagram placed on meter front panel. Examples of connection diagrams of the meter are presented in the [Fig. 4.1](#). Lead wires should be selected based on meter maximum current.



a) DC meters, IEC connection



b) DC meters, ANSI connection



c) TC meters, IEC connection

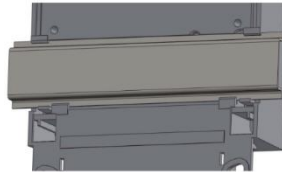
**Fig. 4.1.** Meter connection diagrams

Extra relay (terminals 13, 14, 15, 16, 18, 20), extension communication module (X2 output of the RS-485 interface), output for other meters (X1 output of the RS-485 interface), RS-485 interfaces and pulse output terminals ( S0<sub>1-</sub>, S0<sub>1+</sub>, S0<sub>2-</sub>, S0<sub>2+</sub>) are optional and are supported in different combinations depending on meter model. Terminals for extra relay connection provide easy and reliable connection of wires using screws. Extension communication module is connected via X2 RS-485 output within the terminal block by "plug-in" principle.

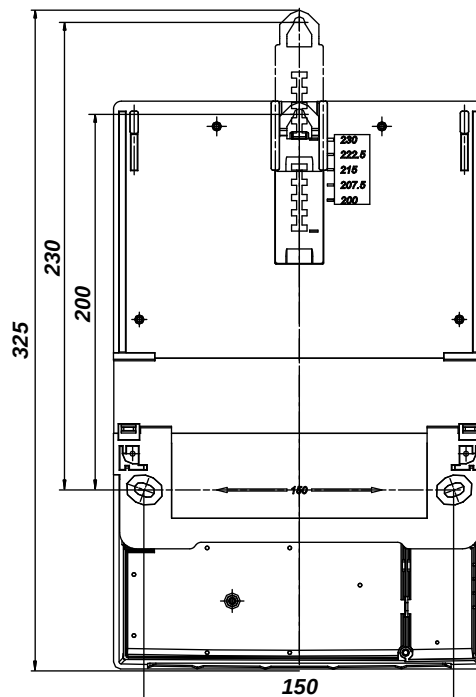
Voltage or current transformers can be used for connection of transformer connected meters.

The meter can be installed in two ways:

- ▶ On DIN-rail:



- ▶ By 3-point fixing. The meter's fixing bar can be easily adjusted on upper screw during installation at the customer's premises.



**Fig. 4.2.** Meter fixing elements for mounting by 3 fixing points

Meter is plug-and-play device, which is registered automatically in the system after its installation and successful communication with a router/data concentrator.

To make sure that the meter is operable check its display after energizing, when the meter is in the test mode\*:

- ✓ All segments are active.
- ✓ The meter firmware version is displayed as **APP X.X.XX**.
- ✓ Screens are displayed according to the meter configuration.

\* For some meter firmware versions the meter display test mode is invoked by pressing a push button.

## 5. DISPLAY

---

The meter supports integrated LCD to cyclically display detailed information on electrical energy consumption. The meter is equipped with a backlit which facilitates the data visualization. The data type, its formats and output sequence are specified in the meter configuration and can be set both locally via the optical port and remotely from the HES. Data identification is presented in the form of symbols and OBIS codes according to IEC 62056-61. Set of display symbols indicate the meter state and may vary for different meter types and versions.

The meter allows automatic or manual screen change mode for data displaying. Automatic mode is intended for a client, manual mode – for service purposes. The list of parameters to be displayed is configured separately for the client and service. The maximum number of parameters in each list is limited to 16.

The meter can display the meter parameters according to preset configuration:

- ✓ date and time energy consumption (in kWh, including total energy by a separate tariff and sum by all tariffs);
- ✓ active power (in kW, total by all phases, and for separate phase);
- ✓ reactive capacitive and inductive power (in kVar);
- ✓ reactive capacitive and inductive energy (in kVarh);
- ✓ instantaneous voltage (in V);
- ✓ instantaneous current (A);
- ✓  $\cos \varphi$  (power factor);
- ✓ meter state;
- ✓ actual tariff;
- ✓ supply network state;
- ✓ basic relay state and reason of disconnection.

In sleep mode, when the power is off, the following fixed list of parameters (by default) can be displayed by push button

- ✓ 1.8.0 (A+);
- ✓ 1.8.1 (A+ rate 1);
- ✓ 1.8.2 (A+ rate 2);
- ✓ 0.9.1 (time);
- ✓ 0.9.2 (date);
- ✓ 0.2.2 (actual tariff plan name);
- ✓ All symbols.

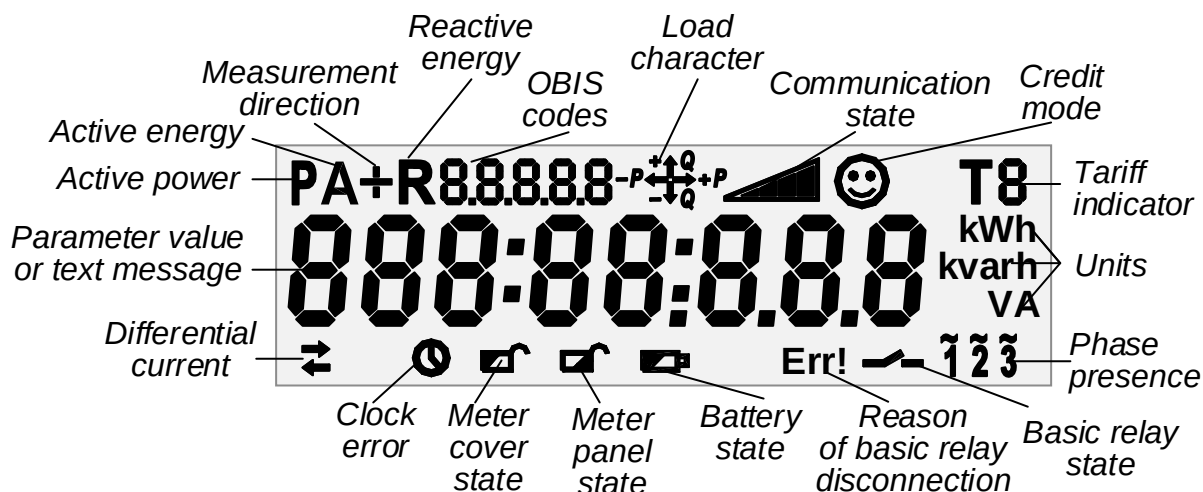
The parameters to be displayed in case of power off, can be negotiated.

The information on the display is divided into 3 lines: Top and bottom lines are parameter indicators, OBIS codes, and flags. OBIS codes allow identification of each measured value. Middle line represents metering data (8 digits of 9.2 mm height, up to 3 decimals, the number of decimals is configured) and measuring units (**kWh**, **kvarh**, **VA**) as well as messages to a consumer\*.

Text message is any text of up to 8 symbols, statically displayed, or a scrolling line of up to 32 symbols, both comprising case-insensitive letters and numbers. Detailed description of text messages form is presented in the [Reference 2](#).

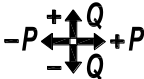
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\* Displayed information depends on the meter version.



**Fig.5.1.** Common symbol and OBIS codes display test mode view.  
All segments are active

**Table 5.1.** Common symbol and OBIS codes display indications

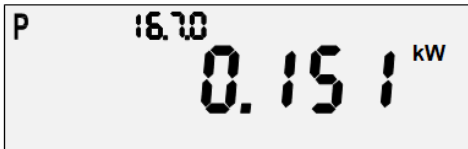
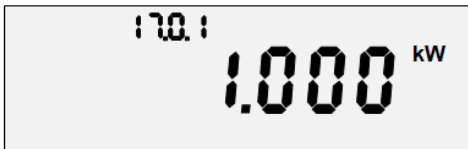
| Indication                                                                          | Description                                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 88.888                                                                              | OBIS (Object Identification System) codes according to IEC 62056-61.                                                                                                                                                                                                                                                                                                                                                                                                    |
|  | Load characteristic, which is used to illustrate energy flow (import/export) and specify metering quadrant.                                                                                                                                                                                                                                                                                                                                                             |
|  | Indicator of communication interface state.<br>1. When PL modem is active, the symbol indicates PLC state (according to the <a href="#">Table 5.2.</a> ).<br>2. When extension communication module (3GPP module) is used for communication upwards through the RS-485 interface (PL modem is not active or not available), the symbol is not displayed. Communication state is indicated using communication module LEDs, available on the meter terminal block cover. |
|  | Meter operates in credit mode.                                                                                                                                                                                                                                                                                                                                                                                                                                          |
| T8                                                                                  | Tariff indicator.                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| 123                                                                                 | Phase presence.<br>Absence of symbol indicates phase voltage absence.<br>Phases blink simultaneously* in case of wrong connection (i.e. wrong sequence of phases or a phase absence).                                                                                                                                                                                                                                                                                   |
|  | Differential current presence (for DC meters only).                                                                                                                                                                                                                                                                                                                                                                                                                     |
|  | Time synchronization error.                                                                                                                                                                                                                                                                                                                                                                                                                                             |
|  | Meter cover is opened.                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|  | Terminal block cover is opened.                                                                                                                                                                                                                                                                                                                                                                                                                                         |

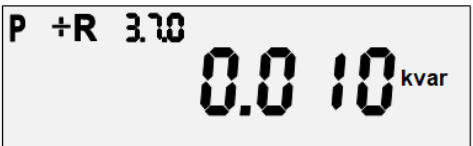
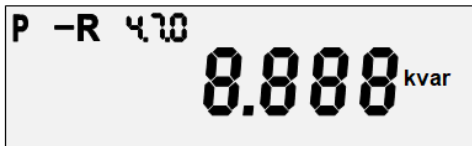
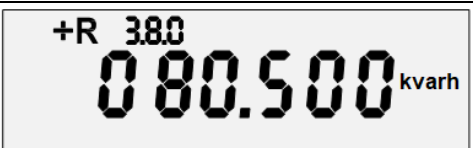
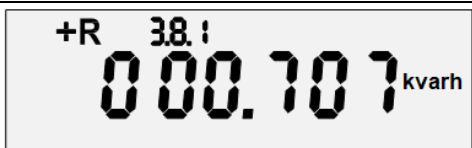
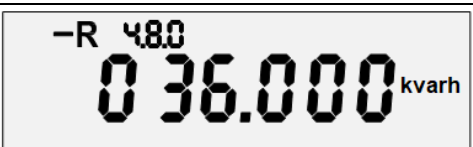
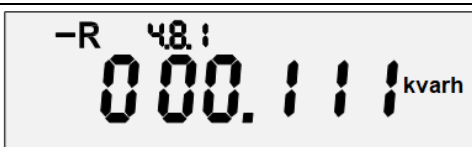
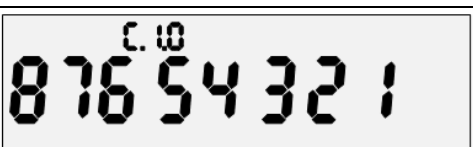
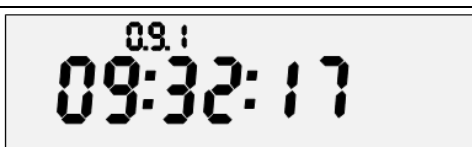
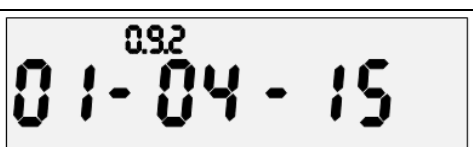
| Indication                                                                        | Description                                                                                                                                                     |
|-----------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  | When battery voltage decreases to 3V from the nominal one of 3.6V, low battery symbol is displayed.<br>When battery is charged enough, symbol is not displayed. |
|  | Basic relay is switched off remotely (for DC meters only).                                                                                                      |
|  | Basic relay is switched off by the reason, e.g. on exceeding of demand or differential current thresholds, by button (for DC meters only).                      |

**Table 5.2.** Indication of PL communication interface state (if PL modem is available)

| Communication indicator   |                                                                                     | PLC modem                        |
|---------------------------|-------------------------------------------------------------------------------------|----------------------------------|
| Blinking full triangle    |    | Modem not configured             |
| Blinking empty triangle   |    | Other error                      |
| Empty triangle            |    | Not registered with DC           |
| Triangle with one mark    |   | Registered – signal level “low”  |
| Triangle with two marks   |  | Registered – signal level “fair” |
| Triangle with three marks |  | Registered – signal level “good” |
| Triangle with four marks  |  | Registered – signal level “high” |

**Table 5.3.** Combination of top line of symbols and readings (examples)

|                                                                                                                                                                  |                                                                                                                                                                   |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Active power (<math>\text{abs}(QI+QIV)-(\text{abs}(QII+QIII))</math>)</p> |  <p>Active power (<math>\text{abs}(QI+QIV)+(\text{abs}(QII+QIII))</math>)</p> |
|  <p>Active power demand limit</p>                                             |  <p>Total Active energy</p>                                                   |
|  <p>Active energy import</p>                                                  |  <p>Active energy import, rate 1</p>                                          |

|                                                                                                                 |                                                                                                                          |
|-----------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------|
|  <p>Active energy export</p>   |  <p>Active energy export, rate 1</p>   |
|  <p>Reactive power import</p>  |  <p>Reactive power export</p>          |
|  <p>Reactive energy import</p> |  <p>Reactive energy import, rate 1</p> |
|  <p>Reactive energy export</p> |  <p>Reactive energy export, rate 1</p> |
|  <p>Device ID</p>            |  <p>Local Time</p>                   |
|  <p>Local Date</p>           |                                                                                                                          |

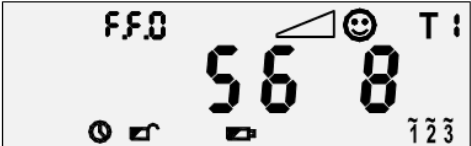
**Table 5.4.** Combination of top and bottom lines of symbols and readings (examples)

|                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                          |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Readings for the import of total active energy (for all tariffs), OBIS code 1.8.0;<br/>no load is applied;<br/>meter PLC modem is registered in HES, "high" signal level is established;<br/>meter operates in credit mode;<br/>current tariff – 1;<br/>all three phases are present;<br/>basic relay is connected, if any;<br/>no flags are set (no alarms are detected)</p> |  <p>Readings for the import of reactive energy at tariff 3, OBIS code 3.8.3;<br/>no load is applied;<br/>meter 3GPP modem is registered in HES, signal level 20-25 is established;<br/>meter operates in credit mode;<br/>current tariff – 6;<br/>supplying voltage is present in the "1" phase;</p> |
|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

|  |                                                                                        |
|--|----------------------------------------------------------------------------------------|
|  | no flags are set (no alarms are detected)<br>no flags are set (no alarms are detected) |
|--|----------------------------------------------------------------------------------------|

**Error Codes** are displayed to identify a number of errors of the Error Object. This object contains itself both common errors (warnings) and critical errors and can be requested locally or remotely to view the errors and alarm events. Additionally, digits from 1 to 8 are displayed on the meter LCD together with the F.F.0 error code (corresponds to the Error Object OBIS code 0.0.97.97.0.255) to identify specified errors. The error codes are displayed in the ascending order from left to right. For each error separate position is fixed. Several error codes may be displayed at the same time. Error codes are displayed together with respective symbols or symbols for other events. By default, error codes are displayed automatically on the LCD among scrolling screens, if proper errors are detected.

**Table 5.5.** Combination of error codes and top and bottom lines of symbols (examples)

|                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                              |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|  <p>Time synchronization error;<br/>low battery voltage;<br/>meter cover is removed;<br/>meter is not registered by the data concentrator;<br/>current tariff – 1;<br/>all three-phases are present</p> |  <p>No phase "1" and "2";<br/>Terminal block cover is removed;<br/>meter PLC modem is registered in HES, "good" signal level is established;<br/>meter operates in credit mode;<br/>current tariff – 3</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

## **6. References**

---

- [1] ADDAX System concepts. Disconnecter. General description
- [2] Messages Handling in ADDAX Devices. Instructions Manual

[Go to Content](#)